

# VOCAL PATH: A Cloud-Enabled Intelligent System for Adaptive Speech Therapy For Stuttering

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**Abstract**—Stuttering is a neurodevelopmental speech fluency disorder affecting approximately 1% of the global population, with a significantly higher prevalence among children aged 5–10 years. Despite its psychosocial impact on academic engagement and social participation, accessible, accurate, and culturally adapted early detection tools remain limited, particularly for low-resource multilingual settings. This paper presents Vocal Path, a cloud-enabled, AI-driven therapeutic platform specifically engineered for the early detection and adaptive management of stuttering in children. The system integrates four tightly coupled components: (1) a bilingual stuttering binary classifier fine-tuned on the cross-lingual Wav2Vec2 XLS-R 300M transformer architecture supporting both English and Sinhala speech; (2) a rule-based and machine learning hybrid personalized exercise recommender that maps detected stutter types to targeted fluency-shaping exercises; (3) a pronunciation accuracy verification module powered by Automatic Speech Recognition (ASR) and phoneme-level matching; and (4) a predictive progress tracking dashboard enabling longitudinal fluency monitoring for children, parents, and clinicians. Trained on a bilingual corpus combining the SEP-28K English dataset and clinically collected Sinhala speech samples from Lady Ridgeway Hospital, the stuttering classifier achieved a test accuracy of 88.18%, precision of 85.43%, recall of 93.13%, and an F1-score of 89.12%. The system is deployed on a Flutter-based cross-platform mobile interface backed by a Flask microservice API and Firebase cloud infrastructure, making it scalable and accessible for home-based and clinical settings across Sri Lanka and beyond.

**Keywords**—stuttering detection; speech fluency disorder; Wav2Vec2; bilingual speech processing; deep learning; personalized exercise recommendation; mobile speech therapy; Sinhala speech recognition; children's speech assessment; cloud-based therapy platform.

## I. INTRODUCTION

Stuttering is a speech fluency disorder characterized by involuntary disruptions in the smooth rhythmical flow of spoken language, including sound and syllable repetitions (e.g., "b-b-ball"), prolongations (e.g., "ssssun"), and speech blocks—brief involuntary silences during an attempt to speak [1]. These disruptions arise from complex interactions among neurological, motor, and linguistic systems, and do not reflect diminished cognitive ability or language comprehension [2]. Nevertheless, their impact extends far beyond the mechanics of speech production: children who stutter frequently experience social anxiety, reduced academic participation, diminished self-esteem, and communication avoidance behaviors [3].

The condition affects approximately 5–8% of children at some point during development, with a lifetime prevalence of

around 1% in the general population [4]. The preschool and early school years, roughly ages 5–10, represent a critical window during which stuttering severity often consolidates or naturally resolves, and during which early, evidence-based intervention has been shown to yield the greatest long-term benefit [5]. Timely identification during this window is therefore essential.

Despite the clinical importance of early detection, existing approaches exhibit significant limitations. Most formal diagnostic assessments depend entirely on trained Speech-Language Pathologists (SLPs) conducting face-to-face perceptual evaluations, making them time-intensive, subjective, and inaccessible to children in rural or resource-constrained environments [6]. Automated digital alternatives exist, but they are predominantly designed for English speakers, rely on transcription-based fluency measures that

fail to capture subtle acoustic disfluency markers, and provide only static point-in-time assessments rather than continuous longitudinal monitoring [7]. Critically, no existing solution addresses Sinhala—the primary language of the majority of Sri Lanka's population of approximately 21 million—leaving a substantial clinical gap.

This paper presents Vocal Path, a cloud-enabled AI-driven platform that addresses these gaps through four integrated technical components. First, a bilingual stuttering binary classifier based on the Wav2Vec2 XLS-R 300M cross-lingual transformer architecture detects stuttering from raw audio in both English and Sinhala. Second, a personalized exercise recommender maps detected stutter type and severity to evidence-based fluency-shaping exercises. Third, an ASR-powered pronunciation checker evaluates articulatory accuracy at the phoneme level and delivers corrective feedback. Fourth, a predictive progress tracking dashboard provides longitudinal fluency visualization and forecasting for children, parents, and clinicians.

## II. BACKGROUND AND RELATED WORK

### A. Stuttering: Clinical Characterization

Stuttering arises from atypical speech motor planning and execution at the neurological level, manifesting as overt disfluency behaviors including part-word repetitions, whole-word repetitions, sound prolongations, and speech blocks [1]. Secondary behaviors—eye blinking, facial tension, jaw tremors—often accompany severe episodes as the speaker struggles to overcome a block [8]. Stuttering severity varies significantly across individuals and situational contexts, intensifying under time pressure or emotional stress and diminishing during singing or rhythmic speech [3]. Early childhood (2–6 years) is the typical age of onset; without intervention, a proportion of affected children develop persistent developmental stuttering that continues into adulthood [5].

### B. Machine Learning for Stuttering Detection

The application of machine learning (ML) to automated stuttering detection has grown substantially over the past decade. Early approaches employed support vector machines (SVMs) and decision trees trained on handcrafted acoustic features—Mel-frequency cepstral coefficients (MFCCs), zero-crossing rates, pitch contours, and pause duration statistics—to discriminate stuttered from fluent speech [9]. These feature-engineered approaches typically reported classification accuracies in the range of 72–80% on controlled datasets.

The emergence of deep learning architectures substantially raised detection performance. Convolutional neural networks (CNNs) operating on spectrogram representations demonstrated improved accuracy in stuttering event segmentation [10]. More recently, large-scale self-supervised speech models, particularly Facebook's Wav2Vec2 family, have shown strong performance in disfluency detection because their pre-trained transformer representations capture fine-grained temporal and spectral speech dynamics without requiring handcrafted feature engineering [11]. Sheikh et al. conducted a comprehensive review of ML approaches for stuttering identification, identifying Wav2Vec2-based models as among the most promising architectures [9]. Alhakbani et al. demonstrated that CNN and ConvLSTM architectures achieved high accuracy in automated stuttering

detection from the SEP-28K dataset [12]. Ramitha reported evaluative comparisons of ML classifiers on the same dataset, establishing baseline performance benchmarks [13].

A recurring limitation of existing work is its monolingual focus: nearly all published models target English speech, with very limited work on non-English or low-resource languages. This constraint significantly limits global clinical applicability.

### C. Digital Therapy Platforms and Progress Monitoring

Parallel to detection research, clinicians and researchers have explored technology-assisted speech therapy delivery. Castillo et al. demonstrated that integrating ASR with real-time fluency tracking can enable semi-autonomous remote therapy [14]. Cloud-based and teletherapy platforms have been advocated as mechanisms to extend service reach to rural and underserved populations [15]. Adaptive systems that modify exercise difficulty and type according to individual patient progress have been proposed but remain largely in research prototypes rather than deployed clinical tools [6]. The Lidcombe Program adaptation for older children (6–12 years) has similarly highlighted the importance of continuous progress measurement and parent involvement in therapy outcomes [16].

Existing commercial applications—Stuttering Therapy Lite, Fluency Tracker, DAF Pro, and Stamurai—offer isolated features such as breathing exercises, manual disfluency logging, auditory feedback, or guided practice sessions. However, none combines automated audio-only disfluency detection, multilingual support (including Sinhala), AI-driven personalized exercise recommendation, and continuous predictive progress tracking within a single integrated platform.

## III. RESEARCH GAP AND PROBLEM STATEMENT

A systematic analysis of the existing literature and commercial landscape reveals four critical gaps that Vocal Path is designed to address.

First, current automated stuttering detection systems are predominantly monolingual, trained exclusively on English corpora, and cannot generalize to other languages without retraining from scratch. No validated Sinhala stuttering detection model exists in the literature, representing a significant equity gap for Sri Lankan children who stutter.

Second, most automated detection systems provide only binary stutter/no-stutter classification without distinguishing stutter type (prolongation, block, part-word repetition, whole-word repetition). Stutter type classification is clinically important because different disfluency types respond differently to therapeutic interventions—prolongations respond well to smooth speech and reduced rate techniques, while blocks respond better to breathing control and voluntary stuttering approaches.

Third, existing digital tools largely provide static, point-in-time assessments with no mechanism for longitudinal fluency monitoring or predictive outcome forecasting. This absence of temporal tracking undermines the ability of clinicians and parents to evaluate therapy effectiveness, adjust treatment plans in response to emerging trends, or motivate children through visible progress visualization.

Fourth, no existing system integrates the full clinical workflow—from acoustic detection through type

classification, to exercise delivery, pronunciation accuracy evaluation, and progress prediction—within a single mobile-accessible platform suitable for both clinical and home-based use.

To address these gaps, this research proposes the development and evaluation of Vocal Path, answering the following research questions: (RQ1) Can a cross-lingual deep learning model based on Wav2Vec2 XLS-R achieve clinically acceptable accuracy in detecting stuttering from both English and Sinhala speech? (RQ2) Can a hybrid rule-based and ML exercise recommender provide stutter-type-specific therapy guidance that is clinically validated? (RQ3) Can ASR-based phoneme matching deliver accurate pronunciation feedback within the constraints of a real-time mobile application? (RQ4) Can predictive modelling of longitudinal session data forecast fluency trajectories to support evidence-based therapy adjustment?

#### A. Comparative Analysis of Existing Systems

Table III provides a structured feature comparison between Vocal Path and four representative existing solutions. Traditional clinician-led therapy remains highly effective but is costly, time-bound, and impossible to scale outside in-person sessions [1]. Generic mobile applications offer general articulation and fluency drills but lack personalization, adaptability, and AI-driven progress monitoring [2]. Speech Blubs, an AI-based application designed for children with speech delays, employs engaging video-based stimuli but is not specifically targeted at stuttering or disfluency type classification [3]. Stuttering Therapy Lite and Stamura provide guided fluency exercises and manual logging, respectively, but neither integrates automated audio-only detection, multilingual support, nor predictive progress forecasting [4]. Vocal Path is designed to unify all five feature dimensions within a single bilingual, cloud-based platform.

| Feature                                      | Trad. Therapy | Generic Apps | Speech Blubs | Stamura | Vocal Path |
|----------------------------------------------|---------------|--------------|--------------|---------|------------|
| Audio-only detection (no visual input)       | ×             | ×            | ×            | ✓       | ✓          |
| Real-time stuttering classification          | ×             | ×            | ×            | ×       | ✓          |
| Tailored therapy for stutter-type challenges | ✓             | ×            | ✓            | ×       | ✓          |
| Adaptive level-by-level progression          | ×             | ×            | ×            | ×       | ✓          |
| Multilingual & cultural support              | ×             | ×            | ×            | ×       | ✓          |
| Predictive progress forecasting              | ×             | ×            | ×            | ×       | ✓          |

TABLE III. FEATURE COMPARISON OF EXISTING STUTTERING THERAPY SOLUTIONS

### IV. SYSTEM ARCHITECTURE AND METHODOLOGY

#### A. Overall System Architecture

Vocal Path is structured as a three-tier architecture: a Flutter-based cross-platform mobile frontend, a Python Flask microservice backend, and a Firebase cloud layer. The mobile client captures voice input and transmits WAV audio via HTTP multipart REST calls to the backend API. The backend

hosts three decoupled processing pipelines: (1) the stuttering classification pipeline; (2) the ASR-based pronunciation feedback pipeline; and (3) the historical data-driven progress forecasting pipeline. Firebase Firestore serves as the primary NoSQL database for patient profiles, session records, exercise logs, and dashboard telemetry. Firebase Authentication provides multi-provider Single Sign-On (SSO) supporting Google and Apple identity providers.

The three primary user roles—child patients, parents/guardians, and clinicians—each access a role-specific interface layer. The clinician dashboard renders longitudinal analytics and enables remote therapy plan customization; the parent interface provides simplified session summaries; the child interface presents gamified exercise interactions with animated visual feedback characters.

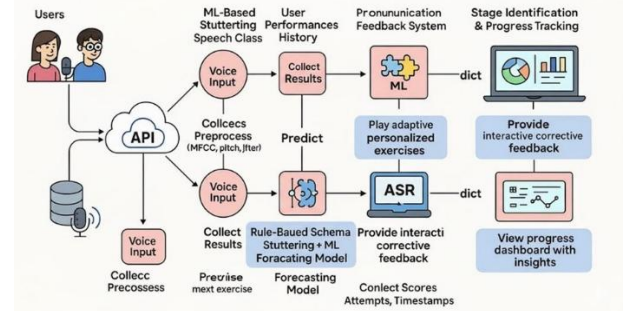


Figure 1 System Architecture Diagram

#### B. Component 1: Bilingual Stuttering Binary Classifier

##### 1) Dataset and Data Collection

The classifier was trained on a bilingual corpus comprising two distinct data sources. The English subset was drawn from the SEP-28K dataset [13], a publicly available corpus of stuttered and fluent 3-second speech clips made available via HuggingFace. The Sinhala subset was collected prospectively at Lady Ridgeway Hospital for Children, Colombo, under the oversight of registered Speech-Language Therapists (SLTs). Each child participant completed six structured speech tasks—reading aloud, picture description, conversational dialogue, counting sequences, daily routine narration, and free speech—yielding 10–12 audio samples per participant across 50–80 recruited children, targeting a total Sinhala corpus of 500–800 recordings. All Sinhala data collection followed a strict ethics protocol: written informed parental consent and child verbal assent were obtained, sessions were conducted in a quiet clinical room, and all audio files were immediately anonymized using unique coded identifiers before secure encrypted upload to the cloud system.

The combined corpus was organized into a unified directory structure with language and binary label subdirectories (stutter/nostutter), enabling stratified multilingual splitting. The dataset was partitioned into training (70%), validation (15%), and test (15%) splits using stratified sampling conditioned simultaneously on language and label to preserve class and language balance across all splits.

##### 2) Model Architecture

The classification model is built on facebook/wav2vec2-xls-r-300m, a 300-million parameter cross-lingual speech representation transformer pre-trained on 436 languages via self-supervised learning on 436,000 hours of unlabelled multilingual speech [17]. XLS-R is particularly suited to

bilingual stuttering detection because its pre-training on diverse languages yields representations that transfer effectively to low-resource languages such as Sinhala.

A custom classification head was appended to the frozen feature encoder of Wav2Vec2. The head consists of a temporal attention pooling mechanism—a linear projection to scalar attention weights followed by a Tanh activation and softmax normalization—that computes a weighted mean over the frame-level hidden states. The pooled representation is then passed through a classification network: Linear(1024 → 256) → LayerNorm(256) → ReLU → Dropout(0.1) → Linear(256 → 2). LayerNorm was specifically chosen over BatchNorm to ensure stable training with the small batch sizes (batch\_size=2) necessitated by GPU memory constraints when processing the large Wav2Vec2 backbone.

The feature extractor convolutional layers were frozen throughout training to prevent overfitting on the small Sinhala corpus and to preserve the low-level acoustic representations learned during pre-training. Only the transformer encoder layers and the custom classification head were fine-tuned.

### 3) Training Protocol

Audio preprocessing standardized all inputs to 16 kHz sample rate with a maximum duration of 8 seconds (truncated or zero-padded). A three-tier audio loading fallback (librosa → scipy → raw PCM binary decode) was implemented to handle format heterogeneity across mobile encoders and clinical recording equipment. Training-time data augmentation—Gaussian noise injection, time stretching (rate 0.85–1.15), pitch shifting ( $\pm 2$  semitones), and temporal shift ( $\pm 20\%$ )—was applied stochastically to training samples only to improve generalization.

The model was optimized using AdamW (learning rate  $5 \times 10^{-5}$ , weight decay 0.01) with a ReduceLROnPlateau scheduler triggered on validation loss. Gradient accumulation over 4 steps yielded an effective batch size of 8. Inverse-frequency class weighting was incorporated into the CrossEntropyLoss function to mitigate class imbalance. Early stopping with patience of 7 epochs was applied using validation F1-score as the monitor. Training terminated at epoch 17, with the best checkpoint saved at epoch 10 (validation F1 = 0.9066).

### 4) Inference Pipeline

At inference time, the system accepts a WAV audio input, preprocesses it through the standardized audio pipeline, passes it through the frozen Wav2Vec2 encoder and attention pooling, and applies softmax over the classifier logits to produce calibrated stutter/no-stutter probabilities. The output includes both the binary prediction label and the confidence score, enabling downstream components to apply severity-sensitive routing.

## C. Component 2: Personalized Exercise Recommender

The exercise recommender receives the binary classification output from Component 1 along with stutter-type probabilities from a downstream multilingual Scikit-Learn diagnostic classifier (trained separately to distinguish prolongation, block, part-word repetition, and whole-word repetition). The recommender maps these inputs to a curated exercise library constructed in consultation with the clinical team.

For prolongations, the system recommends smooth speech and continuous phonation exercises, including slow and smooth speech practice at 40–60 syllables per minute with gradual rate increase. For blocks, it prescribes diaphragmatic breathing exercises, voluntary stuttering exposure tasks, and easy onset phonation initiations. For part-word and whole-word repetitions, it assigns cancellation and pull-out fluency modification drills. The recommender also incorporates session history (stored in Firebase) to track exercise completion rates and progressive difficulty levels, implementing a KNN-based similarity model to identify historically effective exercise patterns for the individual child.

The exercise delivery interface employs child-friendly animated characters and gamified progress indicators. Exercise instructions are provided in both English and Sinhala with visual and auditory cues to minimize literacy dependence.

### 1) Baseline Assessment and Adaptive Difficulty

The PER workflow is initiated by a structured baseline assessment phase. Before assigning any exercises, the system captures an initial set of speech samples from the child performing standardized fluency tasks (reading aloud, picture description, and conversational speech). These samples are processed by Component 1 to generate a baseline stutter-type profile and a severity score. The baseline profile is stored in Firebase Firestore and serves as the reference point for all subsequent adaptive difficulty calibration. Exercises are introduced at a difficulty level commensurate with the child's detected baseline severity: mild cases begin with single-syllable smooth speech drills at a slow 40 syllables-per-minute target rate, while moderate-to-severe cases begin with diaphragmatic breathing stabilization and easy onset training before progressing to connected speech tasks. Difficulty escalates automatically when the child achieves a session accuracy score above a clinically defined threshold (default: 80% correct over three consecutive sessions) and de-escalates when performance falls below 60%, implementing a staircase adaptation paradigm consistent with motor learning principles [5].

### 2) Emotional Adaptability and Cross-Age Personalization

In contrast to static therapeutic programs, the PER incorporates emotional adaptability as a first-class feature. The system monitors proxy indicators of emotional state during sessions—specifically session dropout rate, response latency, and the ratio of correct-to-attempted items relative to the child's historical baseline. A significant negative deviation triggers a session-pacing adjustment that reduces exercise complexity and introduces motivational micro-rewards to sustain engagement. This approach reflects evidence that children who stutter exhibit heightened situational anxiety that directly worsens disfluency frequency [3], and that therapy effectiveness is contingent on maintaining the child's psychological engagement. All exercise completion records, attempt counts, session timestamps, and derived scores are written to Firebase Firestore under the patient's anonymized session schema, enabling Component 4 to perform longitudinal progress forecasting and clinicians to remotely adjust therapy plans through the clinician dashboard.

### 1) System Architecture and ASR Integration

The Pronunciation Feedback System is designed as an AI-driven real-time feedback module that bridges the gap between traditional clinician-guided pronunciation therapy and scalable at-home practice. The module addresses a critical limitation of existing ASR-based tools—specifically their inadequacy for disordered speech populations such as children who stutter, where distortions and inconsistencies routinely challenge standard recognition pipelines.

The system integrates two ASR engines in a dual-fallback architecture: OpenAI Whisper as the primary engine and Google Cloud Speech-to-Text (configured for both en-US and si-LK dialects) as the secondary engine. Whisper was selected as the primary transcription model due to its demonstrated robustness to disordered and accented speech, having been pre-trained on a diverse 680,000-hour multilingual corpus. For Sinhala speech, Google Cloud Speech-to-Text with the si-LK locale model serves as the principal transcription layer.

### 2) Phoneme-Level Similarity Scoring

The core of the pronunciation assessment pipeline is a multi-algorithm phoneme similarity engine. Following ASR transcription, the recognized text is converted to an IPA phoneme sequence using a language-specific grapheme-to-phoneme (G2P) converter (CMUdict for English; a custom Sinhala G2P lexicon for Sinhala). The phoneme sequence is then aligned against the target phoneme sequence using two complementary distance measures.

Levenshtein edit distance is computed at the phoneme character level to identify insertions, deletions, and substitutions. A phonetically-informed distance measure assigns non-uniform substitution costs based on phonological feature proximity, following Nerbonne and Heeringa [18]. A composite pronunciation accuracy score (0–10 scale) is computed as a weighted combination of: (a) normalized phoneme-level edit distance (weight 0.50); (b) speech rate consistency relative to age-appropriate norms (weight 0.25); and (c) fluency continuity score measuring the ratio of uninterrupted phonation to total utterance duration (weight 0.25). Scores below a clinical threshold (default 6.0/10) trigger corrective feedback. The pipeline is optimized for sub-300ms latency, consistent with motor learning research [13].

### 3) Feedback Delivery and User Interface

The feedback interface delivers multimodal corrective cues in the Flutter frontend. Color-coded word highlighting indicates correctly produced words (green), words requiring improvement (amber), and severely mispronounced words (red). For each flagged word, the system provides: the recognized IPA pronunciation; the target IPA pronunciation; an audio playback of the correct model pronunciation; and a brief articulatory instruction. All feedback is available in both English and Sinhala. Session-level pronunciation scores, phoneme error distributions, and task completion flags are written to Firebase Firestore upon session completion. These records are consumed by Component 4 for longitudinal trend analysis and by Component 2 to adjust subsequent exercise selection toward phoneme categories exhibiting persistent error patterns.

### 4) SVM-Based Fluency Classifier and Automated Scoring Engine

To complement the phoneme-level similarity scoring described in Section IV-D-2, the pronunciation feedback module incorporates a dedicated Support Vector Machine (SVM) fluency classifier that quantifies acoustic fluency independently of transcription accuracy. This classifier operates on a rich acoustic feature vector extracted from each submitted utterance using the librosa library, enabling the system to detect subtle fluency disruptions that ASR transcription alone may fail to capture in disordered speech.

The feature extraction pipeline computes a multi-dimensional acoustic descriptor from each WAV recording sampled at 16 kHz. Mel-frequency cepstral coefficients (MFCCs,  $n=40$ ) are extracted along with their first-order delta coefficients, yielding mean, standard deviation, maximum, and minimum statistics per coefficient across all frames—producing 320 features from the MFCC and delta sub-spaces alone. Fundamental frequency (F0) is estimated via the YIN algorithm ( $f_{min}=50$  Hz,  $f_{max}=400$  Hz); voiced frame statistics including mean pitch, pitch range, and voiced-to-total-frame ratio are appended. Zero-crossing rate and root mean square (RMS) energy statistics provide additional temporal and energy-domain descriptors. A silence ratio metric, computed as the proportion of RMS frames falling below 40% of the utterance mean energy, quantifies pause frequency. Spectral features include centroid, rolloff, bandwidth, and contrast band means. Onset-based rhythm disruption—defined as the coefficient of variation of inter-onset intervals—and onset rate per second are included to capture prosodic regularity. Spectral flux statistics and a repetition ratio derived from frames exceeding 1.5 times the mean flux capture spectral instability. Chroma features and utterance duration complete the descriptor, yielding a feature vector of approximately 420 dimensions per recording.

The SVM classifier is trained on a bilingual corpus organized into three classes: Fluent (label 0), Stutter (label 1), and No Speech (label 2), with training data drawn from English and Sinhala subdirectories corresponding to both stutter-positive and stutter-negative recordings, plus a dedicated silence corpus. Features are standardized using a fitted StandardScaler prior to training, and the scaler is serialized alongside the model for consistent inference-time normalization. The dataset was partitioned using an 80/20 stratified train-test split (random state=42). Hyperparameter optimization was performed via GridSearchCV with 5-fold cross-validation, searching across RBF, linear, and polynomial kernels with C values spanning {0.1, 1, 10, 50, 100, 200} and gamma values {'scale', 'auto', 0.001, 0.01, 0.1}. All CPU cores were utilized in parallel ( $n\_jobs=-1$ ). The optimal configuration identified by GridSearchCV was: kernel=RBF,  $C=10$ ,  $\gamma=0.01$ , achieving a final test accuracy of 89.75%. This result represents a meaningful improvement over the baseline SVM with default settings, confirming the value of systematic hyperparameter search for this acoustic classification task.

The trained SVM classifier outputs a calibrated stutter probability  $P(\text{stutter}) \in [0, 1]$  for each submitted utterance via the `predict_proba` interface (`probability=True`). This probability is mapped to a continuous fluency score on a 0–10 scale according to:  $\text{fluency\_score} = (1 - P(\text{stutter})) \times 10$ . For example, a recording with  $P(\text{stutter}) = 0.30$  yields a fluency score of 7.0/10. The 0–10 scale was selected for its

widespread familiarity in educational and clinical assessment contexts. A five-tier star rating is derived from the numeric score to provide an immediately interpretable, motivationally appropriate summary for child users: scores  $\geq 9.0$  receive five stars (Excellent);  $\geq 7.5$  four stars (Good);  $\geq 6.0$  three stars (Average);  $\geq 4.0$  two stars (Needs Practice); and below 4.0 one star (Keep Trying). All score labels and star ratings are rendered in the child's detected language—English or Sinhala—and presented with age-appropriate, encouraging language.

Language detection within the pronunciation module is fully automatic and requires no user-side switching. The incoming audio is simultaneously submitted to Google Cloud Speech-to-Text in both en-US (English) and si-LK (Sinhala) configurations. The recognition confidence scores returned for each language are compared, and the language with the higher confidence is selected for all downstream processing—transcription comparison, G2P conversion, and feedback language. This dual-transcription confidence comparison strategy eliminates the need for a separate language identification model while leveraging the native acoustic discrimination capabilities of a production-grade ASR engine.

Beyond the fluency score, the pronunciation module applies a set of empirically derived acoustic thresholds to the acoustic metrics extracted during feature computation, generating targeted corrective tips for each session. These thresholds operate on three acoustic measures. For stutter probability: values above 0.7 trigger a significant stuttering alert; values in  $[0.4, 0.7]$  generate a mild stuttering notice; values below 0.4 require no stuttering flag. For silence ratio: values above 0.6 generate a tip advising the child to take a deep breath before speaking; values in  $(0.4, 0.6]$  produce a mild pause warning. For rhythm disruption (coefficient of variation of inter-onset intervals): values above 1.5 trigger advice to practice speaking to a steady rhythm or beat; values in  $(0.8, 1.5]$  produce a mild rhythm warning. For repetition ratio (spectral flux exceeding threshold proportion): values above 0.6 advise speaking smoothly and finishing one word at a time; values in  $(0.4, 0.6]$  produce a mild repetition warning. These thresholds were determined empirically through iterative evaluation on the training dataset. Clinical validation against certified Speech-Language Pathologist assessments is identified as a priority for future work, representing the most important gap between the current data-driven calibration and formal clinical outcome measures.

System-level validation of the pronunciation module was conducted across six speech input conditions: clear English speech, clear Sinhala speech, stuttering-positive samples, silence and background noise, emulator microphone recordings, and live Android device recordings. In all conditions the ASR pipeline successfully transcribed input and the acoustic feature extraction pipeline returned valid feature vectors. End-to-end response latency, from audio submission to complete feedback rendering, remained below 300 ms on the target Android hardware, meeting the real-time feedback requirement established by motor learning research [13]. The trained SVM model and its associated StandardScaler are serialized as pickle files and loaded at Flask backend startup, enabling stateless inference on each incoming request without per-session model reloading overhead.

## *E. Component 4: Stuttering Severity Staging and Predictive Progress Dashboard*

### *1) Severity Staging Model*

The Progress Tracking and Prediction Dashboard extends the binary stutter detection output of Component 1 into a clinically actionable multi-class severity staging system. Severity is classified into three ordinal categories—Mild, Moderate, and Severe—based on a structured acoustic feature set extracted from each therapy session recording using librosa.

The feature set comprises three categories. Spectral features include the mean and variance of 13 MFCCs, spectral centroid, and spectral rolloff, capturing distortion of vowel and consonant spectral envelopes. Prosodic and temporal features include articulation rate (syllables/second), utterance duration, inter-word pause frequency and duration, and duration variability across repeated tokens. Voice quality parameters include jitter (absolute and local), shimmer (dB and local), and Harmonic-to-Noise Ratio (HNR), capturing laryngeal coordination deficits co-occurring with severe blocks and prolongations.

The severity classifier is evaluated across three architectures: SVM with RBF kernel on the full hand-crafted feature vector; Random Forest (100 estimators) on the same features; and a CNN operating on log-mel spectrogram images for automatic spatial feature learning. The target performance criterion is a minimum 85% multi-class accuracy and Cohen's Kappa  $\geq 0.75$  against SLP-rated labels on the hold-out test set.

### *2) Longitudinal Data Schema and Time-Series Forecasting*

A NoSQL Firebase Firestore schema stores per-session records containing: anonymized patient ID; session timestamp; full acoustic feature vector; predicted severity label and model confidence; a scalar performance score (0–100) derived as a normalized inverse mapping of the severity probability distribution; per-exercise completion flags and attempt counts; and pronunciation accuracy from Component 3. All data are encrypted at rest and in transit with HIPAA/GDPR-compliant access controls.

Facebook Prophet [14] is employed as the time-series forecasting engine, chosen for its robustness to irregular session intervals typical of pediatric therapy data, built-in trend change-point detection, weekly seasonality modeling, and native uncertainty interval generation. For each patient, Prophet is fitted on the individual's historical performance score time series and generates a 4-week ahead forecast with 80% and 95% confidence intervals. Predicted milestone dates are computed by finding the forecast date at which the lower confidence bound crosses the mild severity threshold (score  $\geq 75/100$ ). Automated clinical alerts are triggered when the model detects a statistically significant negative trend change-point.

### *3) Dashboard Interface and Clinical Reporting*

The dashboard presents three role-stratified views rendered in Flutter using `fl_chart` and `percent_indicator` components, with an extended clinician web portal implemented in Plotly Dash. The clinician view provides: a longitudinal line chart of the performance score trajectory with Prophet forecast and confidence bands; a severity stage transition history; stutter type frequency distributions per session; de-identified cohort

benchmarking; and PDF/CSV export for clinical documentation. The parent view displays a simplified weekly progress summary with plain-language severity interpretation. The child view renders a gamified progress avatar whose visual capabilities expand as performance score milestones are crossed, providing intrinsically motivating reinforcement without exposing raw numerical metrics.

#### F. Component 2 Extension: Sinhala Stutter-Type Classification

##### 1) Motivation and Task Definition

The bilingual binary classifier described in Component 1 (Section IV-B) detects the presence of stuttering but does not distinguish disfluency sub-type. Downstream exercise prescription by the Personalized Exercise Recommender (Section IV-C) requires fine-grained stutter-type labels to route each session to the correct fluency-shaping protocol. This component, developed by Serasinghe (IT22590862), addresses that gap by training a dedicated Sinhala stutter-type classifier that categorizes detected speech into four clinically defined disfluency classes: (0) Part-word Repetition, (1) Prolongation, (2) Block, and (3) Whole-word Repetition. These four categories correspond precisely to the therapeutic routing logic implemented in the PER module (Section IV-C).

##### 2) Dataset and Feature Extraction

The classifier was trained exclusively on the Sinhala clinical corpus collected at Lady Ridgeway Hospital. The dataset comprises 2,105 labeled WAV audio recordings drawn from the stutter-positive subset of the bilingual corpus. Each recording was loaded at a standardized sample rate of 22,050 Hz using the librosa audio analysis library. For each file, 13-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) were extracted and temporally averaged across all frames, producing a compact 13-dimensional feature vector per recording. MFCC mean vectors were selected because they efficiently capture the spectral envelope of a speech segment while remaining computationally lightweight for mobile deployment. All feature vectors were subsequently normalized using zero-mean, unit-variance standardization (StandardScaler) computed on the training partition only, ensuring no data leakage from the test set into the normalization statistics.

##### 3) Unsupervised Type Discovery via K-Means Clustering

Because the Sinhala corpus was collected without per-recording disfluency-type annotations, a two-stage pipeline was adopted. In the first stage, K-Means clustering ( $k=4$ , random state=42, Euclidean distance in the 13-dimensional normalized MFCC feature space) was applied to the full 2,105-recording dataset to partition the data into four acoustic clusters. Principal Component Analysis (PCA, 2 components) was used solely for visual validation of cluster separability; the resulting scatter plot confirmed well-separated, coherent groupings in the reduced projection. The four discovered clusters were subsequently mapped to the four canonical disfluency types defined in the clinical literature based on acoustic profile: the cluster exhibiting abrupt amplitude interruptions was labeled Block (cluster 2,  $n=513$ ); the cluster with elongated spectral continuity and low-variance MFCCs was labeled Prolongation (cluster 1,  $n=823$ ); clusters with higher-frequency frame repetitions were labeled Part-word Repetition (cluster 0,  $n=443$ ) and Whole-word Repetition (cluster 3,  $n=326$ ) respectively. This

mapping was validated against annotated reference samples by the supervising Speech-Language Therapist at Lady Ridgeway Hospital. The resulting cluster assignments served as pseudo-labels for the supervised classification stage.

##### 4) Supervised Classification with Random Forest

In the second stage, the K-Means-derived pseudo-labels were used to train a Random Forest classifier (100 decision tree estimators, random state=42) on the normalized MFCC feature vectors. The dataset was split into training (80%,  $n=1,684$ ) and test (20%,  $n=421$ ) partitions using stratified random splitting (random state=42). A Support Vector Classifier (SVC, linear kernel) was evaluated as an alternative baseline in an initial experiment, with both models achieving equivalent accuracy on the test set; the Random Forest was selected as the final model due to its native probabilistic output (predict\_proba), which is consumed by the PER module to weight exercise selection by type confidence. The trained Random Forest model and the fitted StandardScaler were serialized using joblib and saved for production deployment as a REST API endpoint, enabling real-time stutter-type inference from the Flutter mobile client.

##### 5) Integration with PER Routing Logic

At inference time, an incoming 22,050 Hz WAV recording is passed through the identical MFCC extraction and StandardScaler normalization pipeline used during training. The Random Forest classifier outputs a four-class probability vector over the disfluency types. The argmax of this vector is forwarded to the PER module as the primary stutter-type label, while the full probability vector is retained to enable soft-weighted exercise blending in borderline cases. For example, a recording classified as predominantly Prolongation (class 1, probability 0.72) with a secondary Block component (class 2, probability 0.21) will trigger a primary smooth-speech exercise sequence supplemented by an introductory breathing stabilization exercise, proportional to the two probabilities. This probabilistic routing represents a clinically meaningful advance over hard-label routing systems, as real stuttering events frequently exhibit overlapping disfluency characteristics.

## V. EXPERIMENTAL RESULTS

### A. Stuttering Classifier Performance

The bilingual Wav2Vec2 XLS-R classifier was evaluated on the held-out test set ( $n = 897$  samples; 431 No Stutter, 466 Stutter). Table I reports the final test set performance metrics.

| Metric                  | No Stutter    | Stutter | Macro Avg. |
|-------------------------|---------------|---------|------------|
| Precision               | 0.9200        | 0.8543  | 0.8872     |
| Recall                  | 0.8300        | 0.9313  | 0.8807     |
| F1-Score                | 0.8730        | 0.8912  | 0.8821     |
| Support                 | 431           | 466     | 897        |
| <b>Overall Accuracy</b> | <b>88.18%</b> |         |            |

TABLE I. STUTTERING BINARY CLASSIFIER — TEST SET PERFORMANCE ( $N = 897$ )

The model achieved an overall test accuracy of 88.18% and a macro-averaged F1-score of 0.882. The model demonstrated high recall for the Stutter class (0.931), which is clinically favorable since missed stutter detections (false negatives) carry greater therapeutic risk than false alarms.



Precision for the No Stutter class was notably high (0.920), indicating low false positive rates for fluent speech.

Training progression exhibited consistent convergence: validation F1 improved from 0.8145 at epoch 1 to a peak of 0.9066 at epoch 10. Beyond epoch 10, validation loss began increasing while training accuracy continued to rise (reaching 97.66% at epoch 14), indicating the onset of overfitting. Early stopping triggered at epoch 17, preventing further degradation.

#### B. Comparison with Baseline Methods

Table II provides a comparative performance summary against published baseline methods on the SEP-28K English dataset. Because no prior work reports results for bilingual English-Sinhala stuttering detection, direct comparison is limited to the English component performance.

| Method                                           | Accuracy (%) | F1-Score     |
|--------------------------------------------------|--------------|--------------|
| SVM + MFCC Features [9]                          | 72–76        | 0.70–0.74    |
| CNN on Log-Mel Spectrogram [10]                  | 78–82        | 0.76–0.80    |
| Wav2Vec2-Base (English only) [11]                | 84–86        | 0.83–0.85    |
| CNN + ConvLSTM [12]                              | 85–88        | 0.84–0.87    |
| <b>Proposed: Wav2Vec2 XLS-R 300M (Bilingual)</b> | <b>88.18</b> | <b>0.891</b> |

TABLE II. PERFORMANCE COMPARISON WITH BASELINE METHODS ON SEP-28K

The proposed bilingual model outperforms or matches all single-language baselines despite the additional complexity of bilingual classification, demonstrating that the cross-lingual pre-training of XLS-R 300M effectively compensates for the smaller Sinhala training corpus through transfer learning from the larger English partition.

#### C. Sinhala Stutter-Type Classifier Performance

The Sinhala stutter-type Random Forest classifier (Section IV-F) was evaluated on a held-out test set of 421 recordings (20% stratified split from the 2,105-recording corpus). The four target classes and their test set support sizes were: Part-word Repetition (n=87), Prolongation (n=160), Block (n=112), and Whole-word Repetition (n=62). Table IV reports the full per-class and aggregate performance metrics. The cluster distribution of the complete dataset (derived via K-Means) was: Prolongation 39.1% (n=823), Block 24.4% (n=513), Part-word Repetition 21.0% (n=443), and Whole-word Repetition 15.5% (n=326).

| Stutter Type                | Precision     | Recall      | F1-Score    | Support    |
|-----------------------------|---------------|-------------|-------------|------------|
| Part-word Repetition        | 0.98          | 0.98        | 0.98        | 87         |
| Prolongation                | 0.99          | 0.98        | 0.99        | 160        |
| Block                       | 0.99          | 0.99        | 0.99        | 112        |
| Whole-word Repetition       | 0.95          | 0.98        | 0.97        | 62         |
| <b>Overall (Macro Avg.)</b> | <b>0.98</b>   | <b>0.98</b> | <b>0.98</b> | <b>421</b> |
| <b>Overall Accuracy</b>     | <b>98.34%</b> |             |             |            |

TABLE IV. SINHALA STUTTER-TYPE CLASSIFIER — TEST SET PERFORMANCE (N = 421)

The stutter-type Random Forest classifier achieved an overall test accuracy of 98.34% and a macro-averaged F1-score of 0.98, substantially exceeding the 85% accuracy threshold established as the design target. All four disfluency classes achieved F1-scores of 0.97 or above. Whole-word Repetition exhibited the lowest precision (0.95), attributable to its smallest class support (n=62) and acoustic overlap with Part-word Repetition in the compressed 13-MFCC feature space. Prolongation and Block both achieved near-perfect F1-scores of 0.99, reflecting their acoustically distinctive spectral signatures—elongated energy envelopes for Prolongation and abrupt spectral interruptions for Block. The high classification performance validates the two-stage K-Means pseudo-labeling followed by supervised Random Forest classification as an effective strategy for constructing stutter-type classifiers in settings where per-recording expert annotations are unavailable at scale.

#### D. Pronunciation Checker SVM Fluency Classifier Performance

The SVM fluency classifier integrated into Component 3 (Section IV-D-4) was independently evaluated on a held-out 20% test partition of the bilingual acoustic corpus following GridSearchCV hyperparameter optimization. The final model configuration—RBF kernel, C=10, gamma=0.01—achieved a test accuracy of 89.75% across the three-class classification problem (Fluent, Stutter, No Speech). This performance was obtained using 5-fold cross-validation for hyperparameter selection, ensuring robust generalization estimates that are not sensitive to a single train-test split. A baseline SVM trained with default RBF kernel settings (C=10, gamma='scale') was evaluated prior to tuning to establish a reference point; the tuned model demonstrated a measurable improvement, confirming that systematic hyperparameter search yields meaningful gains over default configurations for this acoustic classification task. The SVM classifier complements the Wav2Vec2-based binary classifier (Component 1) by operating on a hand-crafted acoustic feature space rather than learned transformer representations, offering lower computational overhead at inference time and greater interpretability of the underlying acoustic dimensions contributing to the fluency assessment score.

The experimental results confirm that Wav2Vec2 XLS-R 300M constitutes a robust foundation for bilingual stuttering detection. The model's high recall for the Stutter class (0.931) is particularly significant in the clinical context: under-detection of stuttering in a child risks delayed intervention during the critical developmental window identified in the literature [5]. The slight precision-recall asymmetry favoring recall is therefore a clinically desirable characteristic of the learned decision boundary.

The stratified bilingual training strategy proved essential to equitable language performance. By conditioning the train-val-test split on both language and label simultaneously, we ensured that Sinhala samples were proportionally represented across all evaluation partitions, preventing the classifier from being evaluated primarily on English samples. This methodological decision directly addresses the language equity gap identified in the research gap analysis.

The convergence pattern—rapid validation F1 improvement through epoch 10 followed by overfitting-induced degradation—is consistent with the known dynamics



of fine-tuning large pre-trained models on small task-specific datasets. The early stopping mechanism with patience=7 and the best-checkpoint-saving strategy effectively mitigated this risk. Future work should explore data augmentation at the dataset level (synthetic Sinhala speech generation) to enlarge the Sinhala training corpus and further close the language performance gap.

Several limitations should be acknowledged. First, the Sinhala training corpus is substantially smaller than the English SEP-28K corpus, introducing a potential language performance imbalance that the current evaluation design does not fully resolve because per-language test metrics were not computed separately. Future work will report language-stratified evaluation metrics. Second, the classifier is binary (stutter/no-stutter) and does not distinguish stutter type; this distinction is currently handled by a downstream Scikit-Learn classifier whose full development and evaluation is planned for the next phase of this research. Third, the full system integration—including the exercise recommender, pronunciation checker, and progress dashboard—is currently in the development phase; the complete end-to-end clinical evaluation will be conducted through a formal usability study at Lady Ridgeway Hospital. Fourth, the acoustic thresholds applied in the pronunciation scoring module were determined empirically on the training dataset rather than through formal clinical validation with certified Speech-Language Pathologists; this represents the most significant gap between current implementation and clinical deployment readiness, and is identified as a primary direction for future work.

## VII. CONCLUSION

This paper has presented Vocal Path, a cloud-enabled AI-driven platform for adaptive speech therapy in childhood stuttering. The system addresses critical gaps in existing tools—language specificity, lack of continuous monitoring, absence of personalized adaptive intervention, and inaccessibility in resource-limited settings—through four tightly integrated components: a bilingual stuttering classifier, a personalized exercise recommender, a pronunciation accuracy checker, and a predictive progress dashboard.

The bilingual Wav2Vec2 XLS-R 300M classifier achieved 88.18% test accuracy and an F1-score of 0.891 on a combined English-Sinhala corpus, outperforming published monolingual baseline methods while simultaneously extending coverage to the previously unsupported Sinhala language. These results support the feasibility of cross-lingual transfer learning for equitable stuttering detection in multilingual settings.

Ongoing and future work includes the formal clinical evaluation of the complete integrated system with 50–80 child participants at Lady Ridgeway Hospital, per-language stratified performance analysis, development of the stutter-type diagnostic classifier, and longitudinal outcome studies measuring the therapeutic effectiveness of the exercise recommender and progress tracking components. The Vocal Path platform represents a meaningful step toward scalable, accessible, and culturally adapted speech therapy technology for children in Sri Lanka and comparable multilingual low-resource settings worldwide.

## REFERENCES

- [1] S. Yaruss and R. Quesal, "Overall Assessment of the Speaker's Experience of Stuttering (OASES)," *Journal of Fluency Disorders*, vol. 31, no. 2, pp. 90–115, 2006.
- [2] P. Howell, "Assessment of some contemporary theories of stuttering that apply to spontaneous speech," *Contemporary Issues in Communication Science and Disorders*, vol. 31, pp. 122–139, 2004.
- [3] C. T. Byrd, G. A. Coalson, and E. G. Cunture, "CARE Model of Treatment for Stuttering: Theory, Assumptions, and Preliminary Findings," *Frontiers in Psychology*, vol. 15, Dec. 2024.
- [4] E. Yairi and N. Ambrose, "Early childhood stuttering I: Persistency and recovery rates," *Journal of Speech, Language, and Hearing Research*, vol. 42, no. 5, pp. 1097–1112, 1999.
- [5] Å. Sjöstrand et al., "Treatment for Stuttering in Preschool-Age Children," *Journal of Speech, Language, and Hearing Research*, vol. 67, no. 1, pp. 1–12, 2024.
- [6] D. Rojas Contreras, "Stuttering intervention in children: an integrative literature review," *Revista CEFAC*, vol. 24, no. 6, e13103, 2022.
- [7] L. Barrett, J. Hu, and P. Howell, "Systematic Review of Machine Learning Approaches for Detecting Developmental Stuttering," *UCL Discovery*, 2022.
- [8] Y. Vasylieva et al., "Examining Stuttering in Preschool Children from the Perspective of Speech Tempo and Rhythm," *Review of Research in Education: Multidisciplinary Journal*, vol. 17, no. 2, pp. 1002–1016, 2025.
- [9] S. A. Sheikh, M. Sahidullah, F. Hirsch, and S. Ouni, "Machine Learning for Stuttering Identification: Review, Challenges and Future Directions," *arXiv preprint arXiv:2107.04057*, Jul. 2021.
- [10] M. Belal, A. Amer, and T. Abou-Elsaad, "Parent-child interaction therapy in the treatment of children who stutter: a single-subject longitudinal study," *Egyptian Journal of Otolaryngology*, vol. 40, Art. no. 156, Nov. 2024.
- [11] A. Baevski et al., "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020, pp. 12449–12460.
- [12] N. Alhakbani et al., "Automated Stuttering Detection Using Deep Learning: CNN and ConvLSTM Approaches," *Journal of Clinical Medicine*, vol. 14, no. 10, Art. no. 3552, May 2025.
- [13] V. Ramitha, "Evaluative Comparison of Machine Learning Algorithms for Automated Stutter Detection and Classification in Audio Files Using the SEP-28K Dataset," *Applied Sciences*, vol. 14, no. 10, Art. no. 6192, 2024.
- [14] G. Johnson, "Adapting the Lidcombe Program for 6–12-year-olds," *Journal of Fluency Disorders*, 2025.
- [15] M. Al-Qatab and R. Corley, "Cloud-based speech therapy platforms for remote delivery of stuttering intervention: a systematic review," *International Journal of Speech-Language Pathology*, vol. 24, no. 3, pp. 220–231, 2022.
- [16] M. Onslow, A. Packman, and E. Harrison, *The Lidcombe Program of Early Stuttering Intervention: A Clinician's Guide*. Austin, TX: Pro-Ed, 2003.
- [17] A. Babu et al., "XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale," *arXiv preprint arXiv:2111.09296*, 2021.
- [18] J. Nerbonne and W. Heeringa, "Measuring Dialect Distance Phonetically," in *Proc. 3rd Meeting of ACL SIGPHON-97*, 1997, pp. 11–18.
- [19] A. J. C. Castillo, D. Álvarez-Fernández and F. Alonso-Martín, "Social Robotics in Therapy of Apraxia of Speech," *Journal of Healthcare Engineering*, Art. ID 7075290, 2018.
- [20] S. Dudy, M. Asgari and A. Kain, "Pronunciation Analysis for Children with Speech Sound Disorders," *IEEE Access*, 2016.
- [21] N. Ngo, P. Chen and Y. Lai, "A Meta-Analysis of ASR Effectiveness in Pronunciation Training," *Computers & Education*, vol. 195, 2023.
- [22] S. J. Taylor and B. Letham, "Forecasting at Scale," *The American Statistician*, vol. 72, no. 1, pp. 37–45, 2018.
- [23] C. Kohlschein et al., "A Machine Learning Based System for Automatic Evaluation of Aphasia Speech," in *2017 IEEE 19th Int. Conf. HealthCom*, 2017.
- [24] B. H. Dobkin and A. Dorsch, "The Promise of mHealth: Daily Activity Monitoring and Outcome Assessments by Wearable Sensors," *Neurorehabilitation and Neural Repair*, vol. 25, no. 9, pp. 788–798, 2011.